HW V

EE 588: Optimization for the information and data sciences

University of Southern California

Assigned on: October 29, 2026 Due date: November 12, 2026

The following exercises:

- *Spectrum analysis with quantized measurements.* A sample is made up of n compounds, in quantities $q_i \geq 0$, for $i = 1, \ldots, n$. Each compound has a (nonnegative) spectrum, which we represent as a vector $s^{(i)} \in \mathbb{R}^m_+$, for $i = 1, \ldots, n$. (Precisely what $s^{(i)}$ means won't matter to us.) The spectrum of the sample is given by $s = \sum_{i=1}^n q_i s^{(i)}$. We can write this more compactly as $s = Sq$, where $S \in \mathbb{R}^{m \times n}$ is a matrix whose columns are $s^{(1)}, \ldots, s^{(n)}$.

  Measurement of the spectrum of the sample gives us an interval for each spectrum value, i.e., $l, u \in \mathbb{R}^m_+$ for which

  $$l_i \leq s_i \leq u_i, \quad i = 1, \ldots, m.$$

  (We don't directly get s.) This occurs, for example, if our measurements are quantized.

  Given l and u (and S), we cannot in general deduce q exactly. Instead, we ask you to do the following. For each compound i, find the range of possible values for q_i consistent with the spectrum measurements. We will denote these ranges as $q_i \in [q_i^{\min}, q_i^{\max}]$. Your job is to find $q_i^{\min}$ and $q_i^{\max}$.

  Note that if $q_i^{\min}$ is large, we can confidently conclude that there is a significant amount of compound i in the sample. If $q_i^{\max}$ is small, we can confidently conclude that there is not much of compound i in the sample.

  1. Explain how to find $q_i^{\min}$ and $q_i^{\max}$, given S, l, and u.
  2. Carry out the method of part (a) for the problem instance given in `spectrum_data.m`. (Executing this file defines the problem data, and plots the compound spectra and measurement bounds.) Plot the minimum and maximum values versus i, using the commented out code in the data file. Report your values for $q_4^{\min}$ and $q_4^{\max}$.

- *Estimating a vector with unknown measurement nonlinearity.* (A specific instance of exercise 7.9 in *Convex Optimization*.) We want to estimate a vector $x \in \mathbb{R}^n$, given some measurements

  $$y_i = \phi(a_i^T x + v_i), \quad i = 1, \ldots, m.$$

  Here $a_i \in \mathbb{R}^n$ are known, v_i are IID $\mathcal{N}(0, \sigma^2)$ random noises, and $\phi : \mathbb{R} \to \mathbb{R}$ is an unknown monotonic increasing function, known to satisfy

  $$\alpha \leq \phi'(u) \leq \beta,$$

  for all u. (Here α and β are known positive constants, with $\alpha < \beta$.) We want to find a maximum likelihood estimate of x and ϕ, given y_i. (We also know a_i, σ, α, and β.)

This sounds like an infinite-dimensional problem, since one of the parameters we are estimating is a function. In fact, we only need to know the m numbers $z_i = \phi^{-1}(y_i)$, $i = 1, \ldots, m$. So by estimating ϕ we really mean estimating the m numbers $z_1, \ldots, z_m$. (These numbers are not arbitrary; they must be consistent with the prior information $\alpha \leq \phi'(u) \leq \beta$ for all u.)

1. Explain how to find a maximum likelihood estimate of x and ϕ (i.e., $z_1, \ldots, z_m$) using convex optimization.
2. Carry out your method on the data given in `nonlin_meas_data.m`, which includes a matrix $A \in \mathbb{R}^{m \times n}$, with rows $a_1^T, \ldots, a_m^T$. Give $\hat{x}_{\mathrm{ml}}$, the maximum likelihood estimate of x. Plot your estimated function $\hat{\phi}_{\mathrm{ml}}$. (You can do this by plotting $(\hat{z}_{\mathrm{ml}})_i$ versus y_i, with y_i on the vertical axis and $(\hat{z}_{\mathrm{ml}})_i$ on the horizontal axis.)

Hint. You can assume the measurements are numbered so that y_i are sorted in nondecreasing order, i.e., $y_1 \leq y_2 \leq \cdots \leq y_m$. (The data given in the problem instance for part (b) is given in this order.)

- *Three-way linear classification.* We are given data

$$x^{(1)}, \ldots, x^{(N)}, \quad y^{(1)}, \ldots, y^{(M)}, \quad z^{(1)}, \ldots, z^{(P)},$$

three nonempty sets of vectors in $\mathbb{R}^n$. We wish to find three affine functions on $\mathbb{R}^n$,

$$f_i(z) = a_i^T z - b_i, \quad i = 1, 2, 3,$$

that satisfy the following properties:

$$\begin{aligned} f_1(x^{(j)}) &> \max\{f_2(x^{(j)}), f_3(x^{(j)})\}, \quad j = 1, \ldots, N, \\ f_2(y^{(j)}) &> \max\{f_1(y^{(j)}), f_3(y^{(j)})\}, \quad j = 1, \ldots, M, \\ f_3(z^{(j)}) &> \max\{f_1(z^{(j)}), f_2(z^{(j)})\}, \quad j = 1, \ldots, P. \end{aligned}$$

In words: f_1 is the largest of the three functions on the x data points, f_2 is the largest of the three functions on the y data points, f_3 is the largest of the three functions on the z data points. We can give a simple geometric interpretation: The functions f_1, f_2, and f_3 partition $\mathbb{R}^n$ into three regions,

$$\begin{aligned} R_1 &= \{z \mid f_1(z) > \max\{f_2(z), f_3(z)\}\}, \\ R_2 &= \{z \mid f_2(z) > \max\{f_1(z), f_3(z)\}\}, \\ R_3 &= \{z \mid f_3(z) > \max\{f_1(z), f_2(z)\}\}, \end{aligned}$$

defined by where each function is the largest of the three. Our goal is to find functions with $x^{(j)} \in R_1$, $y^{(j)} \in R_2$, and $z^{(j)} \in R_3$.

Pose this as a convex optimization problem. You may not use strict inequalities in your formulation.

Solve the specific instance of the 3-way separation problem given in `sep3way_data.m`, with the columns of the matrices `X`, `Y` and `Z` giving the $x^{(j)}$, $j = 1, \ldots, N$, $y^{(j)}$, $j = 1, \ldots, M$ and $z^{(j)}$, $j = 1, \ldots, P$. To save you the trouble of plotting data points and separation boundaries, we have included the plotting code in `sep3way_data.m`. (Note that `a1`, `a2`, `a3`, `b1` and `b2` contain arbitrary numbers; you should compute the correct values using CVX.)